

Reaction Mechanisms and Catalysis

Elementary steps, catalysts and enzymes

Name _____ Class _____ Date _____

KEY RELATIONSHIPS

Elementary step: rate law follows the coefficients ($A + B \rightarrow \dots$: rate = $k[A][B]$)

Pre-equilibrium: fast $A + B \rightleftharpoons I$ gives $[I] = K_1[A][B]$

Michaelis-Menten: $v = V_{\max}[S] / (K_m + [S])$ · turnover number $k_{\text{cat}} = V_{\max} / [E]_{\text{total}}$

Enzyme data: $V_{\max} = 150 \mu\text{mol L}^{-1} \text{min}^{-1}$ and $K_m = 0.50 \text{ mmol/L}$ unless stated. Show units in every calculation step.

A. Elementary steps

1. Give the molecularity and the rate law of each elementary step: (a) $\text{O}_3 \rightarrow \text{O}_2 + \text{O}$ (b) $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$ (c) $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$

2. Hydrogen peroxide decomposes in the presence of iodide ions: step 1 (slow) $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$; step 2 (fast) $\text{H}_2\text{O}_2 + \text{IO}^- \rightarrow \text{H}_2\text{O} + \text{O}_2 + \text{I}^-$. (a) Write the overall equation. (b) Identify the catalyst and the intermediate. (c) Write the rate law. (d) Is this homogeneous or heterogeneous catalysis?

B. Testing mechanisms

3. For $2\text{NO} + \text{O}_2 \longrightarrow 2\text{NO}_2$, rate = $k[\text{NO}]^2[\text{O}_2]$. Mechanism A is one termolecular step. Mechanism B is $2\text{NO} \rightleftharpoons \text{N}_2\text{O}_2$ (fast), then $\text{N}_2\text{O}_2 + \text{O}_2 \longrightarrow 2\text{NO}_2$ (slow). (a) Show that both mechanisms give the experimental rate law. (b) Which is more likely, and why?

4. Ozone decomposes, $2\text{O}_3 \longrightarrow 3\text{O}_2$, by: $\text{O}_3 \rightleftharpoons \text{O}_2 + \text{O}$ (fast), then $\text{O} + \text{O}_3 \longrightarrow 2\text{O}_2$ (slow). Derive the rate law, and explain why adding O_2 slows the reaction.

C. Catalysts and enzymes

5. Ethene reacts with hydrogen on a nickel catalyst. Describe the four stages of the catalysed reaction, and explain why the nickel is used as a fine powder.

6. Using the enzyme data, calculate the rate at (a) $[\text{S}] = 0.50 \text{ mmol/L}$ (b) $[\text{S}] = 4.5 \text{ mmol/L}$. (c) At what $[\text{S}]$ is the rate 90 % of V_{max} ? (d) At $[\text{S}] = 0.050 \text{ mmol/L}$, compare the exact rate with the low- $[\text{S}]$ approximation $v \approx (V_{\text{max}}/K_{\text{m}})[\text{S}]$.

D. Challenge

7. Two inhibitors are tested at $[S] = 1.0 \text{ mmol/L}$. Inhibitor X is competitive (apparent $K_m = 1.5 \text{ mmol/L}$, $V_{\max}$ unchanged). Inhibitor Y is non-competitive ($V_{\max}$ falls to $90 \mu\text{mol L}^{-1} \text{ min}^{-1}$, K_m unchanged). (a) Calculate the rate with no inhibitor, with X and with Y. (b) Explain how measuring the rates at $[S] = 10 \text{ mmol/L}$ would tell the two inhibitors apart.

8. The enzyme concentration in the experiment is $0.025 \mu\text{mol/L}$. (a) Calculate the turnover number in s^{-1} . (b) Explain why the rate of the enzyme reaction first rises and then falls as the temperature is raised from 20°C to 70°C .